

Original Article

New Zealand experience of I¹²⁵ brachytherapy for choroidal melanoma

Rebecca Stack MBChB,¹ Mark Elder FRANZCO,^{1,2} Al Abdelaal FRAR,³ Rudy Hidajat PhD^{1,4} and Richard Clemett FRANZCO¹

¹Ophthalmology Department, ³Oncology Department and ⁴Department of Medical Physics and Bioengineering, Christchurch Hospital, and

²Christchurch School of Medicine, Otago University, Christchurch, New Zealand

ABSTRACT

Purpose: To examine patient survival, visual function and complications in all patients with choroidal melanoma treated with I¹²⁵ brachytherapy between 1995 and 2003 at the authors' institution. To compare the results from their institution with those from international series.

Methods: Data were collected on 92 consecutive patients. Dosimetry was calculated for the lens, fovea, optic nerve and tumour apex. Follow-up status of each patient domiciled outside the treatment centre was analysed from a postal questionnaire.

Results: Average pretreatment tumour dimensions were 3.9 mm (thickness) and 15.2 mm (diameter). Complete follow-up data were available on 92% of patients. Regression of tumour occurred in 88% of cases. Five enucleations were performed. There were five melanoma-related deaths. Visual acuity remained >6/12 in 35% of patients and >6/60 in 51% of patients. The most frequently observed complication was radiation retinopathy (maculopathy 23%, peripheral retinopathy 17%). Radiation cataract was seen in 11% and optic neuropathy in 10%. Dose of more than 90 Gy to the macula gave a 63% chance of developing maculopathy ($P < 0.01$). A tumour larger than 4 mm significantly increased the risk of developing radiation maculopathy ($P = 0.003$). Development of radiation cataract was dose-related; >25 Gy to the lens gave a 44% risk of cataract development ($P < 0.001$). For tumours less than 4 mm from the disc margin there was a 50% risk of optic neuropathy ($P < 0.001$).

Conclusions: Patient outcomes following brachytherapy were excellent with a high percentage of patients retaining mobility vision. Development of complications was related to the tumour location and dose to non-tumour structures.

Key words: brachytherapy, choroidal melanoma, iodine, plaque.

INTRODUCTION

Plaque radiotherapy has become an established method for the treatment of choroidal melanoma. The Collaborative Ocular Melanoma Study (COMS) confirmed equal survival rates for patients with choroidal melanoma with either plaque radiotherapy or enucleation.¹ Plaque radiotherapy has the added benefit of retention of the eye for cosmesis and in many cases retention of good visual acuity.

Christchurch was the only New Zealand centre where ocular I¹²⁵ plaque radiotherapy was available in 1995–2003. The reported patients represent the complete New Zealand population treated with this modality within the study time frame.

This study aimed to determine outcomes for I¹²⁵ plaque radiotherapy used to treat choroidal melanoma within New Zealand. By examining patient outcomes by survival, visual outcome and development of complications, and comparing the dose of radiotherapy each patient received, we aimed to find relationships between tumour characteristics, treatment parameters and outcomes. We also wanted to compare our outcomes with those reported in the literature from other major international tumour centres.

METHODS

All patients who had choroidal melanoma treated with radioactive I¹²⁵ plaques between 17 November 1995 and 7 April 2003 in Christchurch were analysed. A retrospective analysis of survival, visual function and the development of complications in all patients with choroidal melanoma treated with I¹²⁵ brachytherapy was performed.

Data were compiled from detailed audit forms completed at the preoperative, operative and postoperative visits. These were all completed by two authors (M.E. and R.C.) who

jointly managed the tumour service and filled in patient records.

All tumours were measured on the same high definition B-scan ultrasound machine (Phillips ATL 5000, Bothell, WA, USA). From multiple ultrasonogram images, tumour thickness and maximum horizontal diameter, distance from tumour to optic disc and tumour to lens centre were measured. Preoperative data included age, sex, systemic illness (diabetes mellitus, hypertension) and best-corrected visual acuity. Tumour data comprised tumour location, tumour pigment and associated retinal findings (retinal detachment, subretinal fluid). Operative information recorded which muscles were removed to allow placement of the plaque, whether the tumour was located with transillumination or indentation and any complications of surgery. Radiation doses to tumour apex, optic disc, fovea, sclera and lens (cGy) were prospectively recorded as well as the radiation rate to the apex of the tumour (cGy/h). These dosimetry calculations were derived using BEBIG (BEBIG Isotopen – und Medizintechnik GmbH, Berlin, Germany) software. A total dose of 100 Gy was administered to each prescription point or tumour apex at a dose rate between 50 and 125 cGy/h. The seeds' activity varied between 3 and 8 mCi depending on the availability of seeds and the time of treatment. Activity of each batch of iodine seeds is within 7% of that stated by the supplier (Amersham Health, Princeton, NJ, USA). All measurements or plans were checked by two physicists and were approved by the principal physicist and oncologist. Final follow up was the last visit to an ophthalmologist. Details were from patient notes if available. Following treatment, patients from outside our area were returned to the referring ophthalmologist for follow up. Follow-up data for this group was derived from a postal questionnaire sent to the ophthalmologist monitoring the patient.

The follow-up questionnaire asked for details on length of follow up, mortality and cause of death (tumour-related or from other cause), evidence of metastatic spread, need for enucleation and reason if it occurred, visual acuity in each eye, measurements of the tumour (by ultrasound if available or clinical impression if not), any further surgery the patient had required and complications that had developed. The complications were grouped in radiation-induced and other.

The major outcomes analysed in this study were the development of radiation events including maculopathy, cataract, optic neuropathy, vitreous haemorrhage, neovascular glaucoma, scleral necrosis. Visual acuity decrement and tumour events including recurrence, metastasis and death were also analysed.

Definitions of complications^{2–5}

- 1 Radiation cataract: any lens opacity not present before radiation.
- 2 Radiation retinopathy: any retinal capillary bed changes (non-perfusion, dilatation, haemorrhages,

microaneurysms), retinal exudation, oedema, vascular sheathing, nerve fibre layer infarctions or neovascularization in peripheral retina.

- 3 Radiation maculopathy: any of the above changes in foveal area affecting central vision.
- 4 Radiation optic neuropathy: peripapillary exudation, haemorrhage, optic disc swelling or atrophy.
- 5 Vitreous haemorrhage: any amount of vitreous blood.
- 6 Neovascular glaucoma: the presence of iris and/or angle new vessels and elevated intraocular pressure.
- 7 Scleral necrosis: thinning of the sclera permitting visualization of uveal tissue.
- 8 Tumour recurrence: any degree of enlargement of the residual tumour in base or thickness as detected by ophthalmoscopy or ultrasonography.

RESULTS

Ninety-two consecutive patients (47 women and 45 men) with choroidal melanomas (48 right and 44 left eyes) were treated with radioactive iodine plaques in our institution between March 1995 and January 2003. Median age was 60.9 years (range 28–83 years). The location of the tumour was superior in 5 (5%), supero-temporal in 22 (24%), infero-temporal in 24 (26%), inferior in 3 (3%), macular in 9 (10%), temporal in 9 (10%), nasal in 6 (7%), supero-nasal in 9 (10%) and infero-nasal in 5 (5%). The pigmentation was described as dense in 34 (39%), moderate in 46 (52%), light in 6 (7%) and amelanotic in 2 (2%). There was no record of pigmentation in the remaining four patients' notes. In 57 patients (65%) there was no retinal detachment. In 20 patients (23%) there was localized detachment, in 8 (9%) there was up to half the retina detached and in 1 (1%) the detachment was subtotal. No comment was made in six cases. Tumours were measured by ultrasound in the week prior to the plaque being applied. The mean thickness was 3.9 mm (range 1.8–8.0 mm), mean diameter was 10.4 mm (range 5.0–15.2 mm), mean proximity to optic disc was 5.0 mm (range 0–15.7 mm) and to lens base was 14.1 mm (range 1.1–18.9 mm). In six eyes (6.5%) the tumour abutted the optic nerve. There was ciliary body involvement in 10 cases (11%).

All patients were treated with radioactive iodine¹²⁵ plaques. In total, 66 15-mm plaques, 23 11-mm plaques and 3 18-mm plaques were used. In 12 eyes notched plaques were used because of proximity of the tumour to the optic nerve. Muscles were removed to allow for accurate plaque placement in 69 cases (75%). The plaques were placed intraoperatively using transillumination in all but one case where indentation was required. There were no intraoperative complications reported. The radiation time and dose to critical ocular structures are given in Table 1. The initial visual acuity was better than 6/60 in 86.2% of patients and better than 6/12 in 49.4% of patients. Details on the final visual acuity and change in visual acuity are given in Table 2.

Follow-up data were available on 84 patients (92.3%), an excellent response rate to a postal survey. The follow-up period was 4–86 months (mean 35 months, median

Table 1. Dosimetry for the 92 patients receiving I¹²⁵ brachytherapy in Christchurch

Variable	Mean $\pm$ SD	Minimum	Maximum
Duration (h)	113.7 $\pm$ 37.4	38.0	235.4
Optic nerve dose (Gy)	46.8 $\pm$ 38.9	7.1	219.8
Lens dose (Gy)	16.8 $\pm$ 13.1	4.3	82.1
Macular dose (Gy)	73.0 $\pm$ 13.1	6.7	331.5
Scleral dose (Gy)	313.4 $\pm$ 57.3	150.4	770.0

The total dose was taken as 100 Gy to the prescription point, which is the apex of the tumour. Dose rates were all between 50 and 125 cGy/h.

Table 2. Visual acuity in our study before ($n = 92$) and after ($n = 84$) plaque therapy

Visual acuity	Affected eyes n (%)	Fellow eyes n (%)
Pre-plaque		
>6/12	52 (56)	NA
>6/60	87 (95)	NA
Post-plaque		
>6/12	29 (35)	NA
>6/60	51 (60)	NA
$\downarrow \geq 2$ lines	62 (74)	4 (5)
$\downarrow \geq 5$ lines	39 (44)	0

NA, not applicable.

31 months). Duration of follow up was >1 year in 18 patients, >2 years in 27 and >5 years in 18. Tumours were monitored for growth by clinical impression alone in 43 patients whereas in 41 patients measurements of the tumour dimensions by ultrasound were available. In 88% of cases radiotherapy was reported to halt growth or cause regression of the tumour. The total number of choroidal melanomas not responsive to radiotherapy and reported as growing, by ultrasound or clinical impression, was 10 (12%). Enucleations were performed in five cases. Three were because of tumour growth, one was for a blind painful eye secondary to neovascular glaucoma and one was because persistent vitreous haemorrhage prevented monitoring of the tumour.

Of the remaining five patients, one patient died of unrelated causes while awaiting review; two patients were found to be stable with smaller tumours on repeat ultrasound measurements at our institution and no intervention required; and two had tumour growth confirmed on ultrasound. One of these refused enucleation and the second was awaiting a decision on further management at the time of this report.

The most frequently observed radiation complication was a maculopathy, which was present in 19 cases (23%). Peripheral radiation retinopathy was seen in 14 cases (17%), radiation cataract in 9 cases (11%), optic neuropathy in 8 cases (10%) and vitreous haemorrhage in 3 cases (3.5%). All of the patients with vitreous haemorrhage had radiation retinopathy. There were no cases of scleral necrosis.

There were five melanoma-related deaths (5.9%). None of these patients had had enucleations. All were noted to

have lesions that had regressed since brachytherapy. It is interesting to note that all of these patients were monitored outside our centre and the follow up was done by clinical impression. None had had repeat ultrasound scanning since operation.

The other four deaths were motor vehicle accident (1), myocardial infarction (1), prostatic cancer (1) and unknown (1). The patient with prostate cancer was diagnosed shortly after brachytherapy and it has been postulated that the lesion was a metastasis rather than a primary melanoma.

DISCUSSION

This study examines the results of treatment of a consecutive series of 92 patients with choroidal melanoma who were treated at one centre in New Zealand with I¹²⁵ brachytherapy. We have compared our results with reports in the literature specifying plaque type concentrating on those using radioactive iodine but including some treatments with ruthenium-106, iridium-192 and cobalt-60 plaques. Major series from the literature were used to benchmark our results: the COMS trial used I¹²⁵ brachytherapy;^{1,6} Shields *et al.* used I¹²⁵ (59% I¹²⁵);⁵ Gunduz *et al.* used a variety of radioactive plaques (but 382/630 I¹²⁵);⁴ and Mameghan *et al.* used I¹²⁵ for the 52 Australian patients they treated.⁷ Tumour dimensions and location are listed in Table 3. There are similarities between size and location within the series compared.

Patient demographics

Patients whom we treated had an average age of 60.9 years and there was no sex bias. Half of Gunduz *et al.*'s 630 patients were female and their median age of 59 years was similar to our patients' mean age of 61 years.⁴ Similar sex distribution and median age occurred in the Australian experience.⁷

The plaque

The commonest plaque size used was 15 mm (70% for Mameghan *et al.*,⁷ 85% for Gunduz *et al.*⁴ and 72% of ours). Details on the duration of treatment and doses to vital ocular structures for all the major studies are given in Table 4. Details to structures other than the tumour were not given for the Mameghan *et al.* group.⁷

Follow up

Our follow-up period was 4–86 months (mean 35 months). This was shorter than the Gunduz *et al.* group (16–224 months, median 63 months)⁴ and the Shields *et al.* group (mean follow up 61 months).⁵ Follow up for the COMS group was longer than 24 months for all patients⁶ whereas the Mameghan *et al.* series had a much shorter mean follow up of only 19 months.⁷ A comparison of visual change in all the major studies is given in Table 5. Although details of visual change were not given, the Mameghan *et al.* group reported vision to remain stable in 40% of patients.⁷ This is

Table 3. Tumour dimensions of our series compared with the major studies in the literature

Tumour	Christchurch	Shields ⁵	COMS ⁶	Gunduz ⁴
Dimensions (mm)				
Average largest diameter	10.4	10	11.5	10
Average thickness	4	4.7	6.2	4
Location (%)				
Superior	5	6		4
Superotemporal	24	22		26
Inferior	3	6		7
Inferotemporal	26	24		31
Temporal	10	11		19
Macula	10	<1	13	15
Nasal	7	5		3
Superonasal	10	12		5
Inferonasal	5	13		5

COMS, the Collaborative Ocular Melanoma Study.

Table 4. The mean dose of radiotherapy for our series compared with the literature

Variable	Christchurch (<i>n</i> = 92)	COMS (median-sized) (<i>n</i> = 623) ⁶	Shields (<i>n</i> = 1300) ⁵	Gunduz (<i>n</i> = 630) ⁴
Duration (h)	113.7		141.2	120.0
Dose rate to apex (cGy/h)	98.6	70.5	80.0	80.0
Apical dose (Gy)	102.2	85.0	95.2	91.2
Optic nerve dose (Gy)	46.8	52.1	70.6	59.0
Lens dose (Gy)	16.8	15.6	24.1	11.7
Macular dose (Gy)	73.0	79.0	86.9	122.1
Scleral dose (Gy)	313.4			

COMS, the Collaborative Ocular Melanoma Study.

Table 5. Visual acuity outcomes in our series compared with the literature

Post-plaque visual acuity	Christchurch (35/12) (%)	COMS (36/12) (%) ⁶	Shields (%) ⁵	Gunduz (%) ⁴
>6/12	35	31		9
>6/60	60	50	49	44
↓ ≥ 2 lines	74			
↓ ≥ 5 lines	44	49	54	

COMS, the Collaborative Ocular Melanoma Study.

similar to the 38% of patients in our sample who remained stable or lost less than two lines of vision.

Outcomes

Tumour shrinkage or cessation of growth was seen in 88% of our patients. In the Gunduz *et al.* sample 91% of tumours responded to the plaque therapy.⁴ Growth was seen in five patients in our series and five other patients died of melanoma-related causes. The status of the tumour is not known in the cases of melanoma-related death. The metastases may have occurred prior to plaque therapy and the local tumour may have responded to treatment. This compared favourably with the study by Gunduz *et al.* where recurrence was seen in 8% of patients and recurrences were treated with enucleations (63%) or a second plaque (37%).⁴

Complications

The complications in our study were similar to those seen in other major international series (Table 6). Radiation retinopathy was the most common complication in our series. It is also the most commonly reported complication in the literature seen in 42% of the Shields *et al.* series.⁵ In our series the peripheral retina was involved in 20% and the macula in 27% of cases. The chance of developing radiation retinopathy in our series was significantly related to dose. If the macula received a dose of 90 Gy or more there was a 63% chance of developing maculopathy ($P < 0.01$). Development of maculopathy was also significantly related to the size of the tumour. Patients with tumours larger than 4.0 mm were significantly more likely to develop radiation maculopathy ($P = 0.003$).

Table 6. Complications in our series compared with the literature

Complication	Christchurch (%)	Shields (%) ⁵	Gunduz (%) ⁴	Mameghan (%) ⁷
Maculopathy	22	43	38	
Retinopathy	17	43		36 [†]
Optic neuropathy	10		12	13
Cataract	11	40	30	8
Vitreous haemorrhage	4	7	9	
Neovascular glaucoma	1	7	8	6
Growth	11		8	10
Enucleation	6	8	5	8
Melanoma death	6	10	11	

[†]Did not differentiate into maculopathy and peripheral retinopathy.

A dose of 25 Gy or more to the lens resulted in a 44% chance of developing cataract in our series ($P < 0.001$). Even at 20 Gy there was a significant risk of cataract in 40% of patients ($P < 0.01$). The incidence of cataract in our series was 13% compared with 30% in the Shields *et al.* group.⁵ This difference may represent a lower average dose of radiation to the lens, 17 Gy in our series compared with an average 24 Gy in their series. The lower average dose may be attributed to the fact that isotopes other than iodine were used. Iodine has less scatter radiation than other isotopes. Their series also had a longer follow-up period giving more time for cataract to develop.

Radiation optic neuropathy was seen in eight cases in our series. There was no significant dose–response relationship or relationship to tumour size, but location of the tumour was a highly significant predictor of optic neuropathy. If the tumour was less than 4 mm from the optic disc margin there was a 50% chance of developing optic neuropathy ($P < 0.001$).

This study has a large series of patients with complete follow up. It represents a series of patients treated by the same clinician over a long duration. It has the disadvantage of follow up being conducted in a variety of centres by multiple clinicians. Conclusions regarding outcomes are therefore based on other observer's comments given by written feedback. The 92% response rate to a written survey is exceptional. Tumour size at follow up was determined in the majority of cases by clinical impression rather than formal ultrasound measurement, which would be more accurate.

In conclusion, the COMS trial established that brachytherapy is an attractive alternative to enucleation for the treatment of choroidal melanomas.¹ The eye is retained and often patients have useful long-term visual function. In this series we have documented the New Zealand experience of I¹²⁵ brachytherapy and compared our results with those from major international centres.

Our visual outcomes and complications are comparable with results from international centres.^{1–7} In two-thirds of cases, even with complications, patients retain vision of 6/60 or better in their tumour eye. One-third of patients have vision of driving standard or better (6/12), in the treatment eye, following plaque therapy.

Complications are determined by the dose of radiation received by vital non-tumour structures. They are higher for large tumours, those close to the macula and those in proximity of the optic disc. From the results of our study predictive information on prognosis after treatment can be given to each patient.

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